

TASK 1 — Improve Pose Detection Accuracy

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1. Introduction

Human pose detection is an important concept in computer vision that involves identifying and tracking the positions of different parts of the human body from images or video streams. The system detects key points, known as landmarks, such as the head, shoulders, elbows, wrists, hips, knees, and ankles. These landmarks are then connected to form a skeletal representation of the human body.

Pose detection has wide applications in areas such as fitness tracking, healthcare monitoring, gaming, gesture recognition, surveillance, and augmented reality systems. With the advancement of artificial intelligence, pose detection systems have become more accurate and capable of real-time performance.

The objective of this task was to improve the accuracy and stability of pose detection using MediaPipe and OpenCV in a real-time environment.

2. Objective

The main objectives of this task are:

- To implement real-time human pose detection using MediaPipe and OpenCV
 - To improve the accuracy of body landmark detection
 - To enhance stability by reducing jitter in landmark positions
 - To integrate segmentation for better visualization
 - To monitor performance using FPS (Frames Per Second)
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3. Tools and Technologies Used

The following tools and technologies were used:

- **Python** — Programming language used for implementation

- **OpenCV** — Used for capturing and processing video frames
 - **MediaPipe** — Used for pose detection and landmark estimation
 - **NumPy** — Used for numerical operations and segmentation processing
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4. System Implementation

The system was implemented to process real-time video from the webcam and detect human pose landmarks efficiently.

The working pipeline of the system includes:

1. Capturing video frames using the webcam
2. Converting frames from BGR to RGB format
3. Passing the frames to the MediaPipe pose model
4. Detecting body landmarks
5. Drawing skeleton connections between landmarks
6. Applying segmentation to separate the subject from the background
7. Displaying the output in real time

This pipeline ensures continuous detection and tracking of human body movements.

5. Code Implementation

The following code was used to implement the improved pose detection system:

```
import cv2
import mediapipe as mp
import time
import numpy as np

mp_pose = mp.solutions.pose

pose = mp_pose.Pose(
    static_image_mode=False,
    model_complexity=2,
    smooth_landmarks=True,
    enable_segmentation=True,
    min_detection_confidence=0.7,
```

```

    min_tracking_confidence=0.7
)

mp_draw = mp.solutions.drawing_utils

cap = cv2.VideoCapture(0)

pTime = 0

while True:
    success, frame = cap.read()
    if not success:
        break

    frame = cv2.resize(frame, (960, 720))
    rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)

    results = pose.process(rgb)

    if results.pose_landmarks:
        mp_draw.draw_landmarks(
            frame,
            results.pose_landmarks,
            mp_pose.POSE_CONNECTIONS,
            mp_draw.DrawingSpec(color=(0,255,0), thickness=2, circle_radius=2),
            mp_draw.DrawingSpec(color=(0,0,255), thickness=2)
        )

    if results.segmentation_mask is not None:
        mask = results.segmentation_mask
        condition = mask > 0.5
        bg = cv2.GaussianBlur(frame, (55, 55), 0)
        frame = np.where(condition[..., None], frame, bg)

    cTime = time.time()
    fps = int(1 / (cTime - pTime)) if cTime != pTime else 0
    pTime = cTime

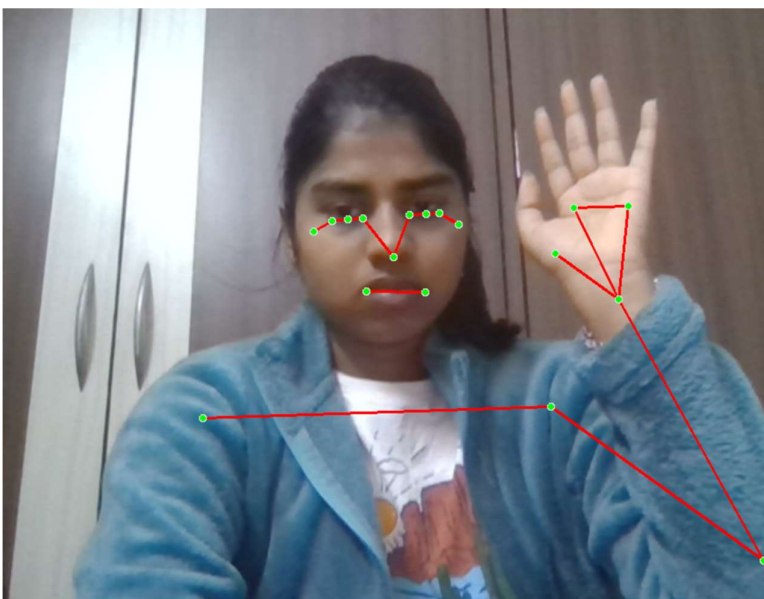
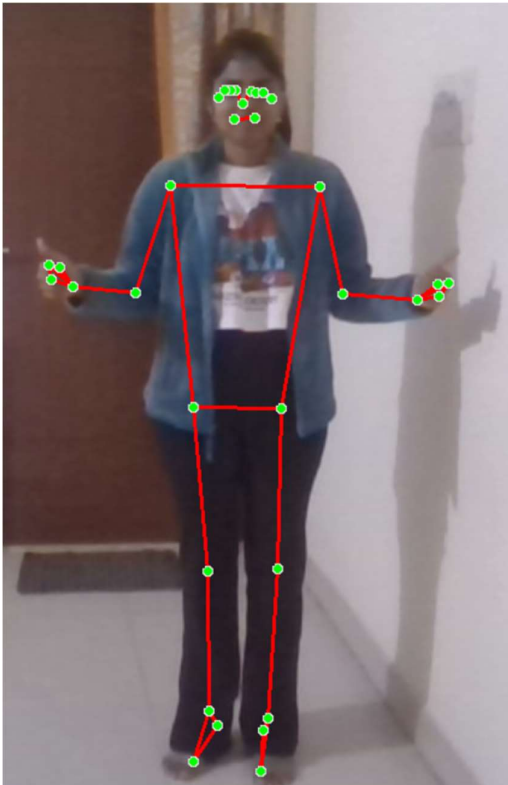
    cv2.putText(frame, f"FPS: {fps}", (10, 30),
                cv2.FONT_HERSHEY_SIMPLEX, 1, (255, 0, 0), 2)

```

```
cv2.imshow("Improved Pose Detection", frame)
```

```
if cv2.waitKey(1) & 0xFF == ord('q'):  
    break
```

```
cap.release()  
cv2.destroyAllWindows()
```



6. Improvements Applied

Several improvements were made to enhance the performance of the pose detection system:

6.1 Model Complexity Enhancement

The model complexity was increased to improve the precision of detecting body landmarks. A higher complexity model provides better accuracy in identifying joint positions.

6.2 Landmark Smoothing

Smoothing was enabled to reduce fluctuations in landmark positions between frames. This ensures stable tracking and minimizes jitter.

6.3 Confidence Threshold Adjustment

Detection and tracking confidence thresholds were increased to ensure that only reliable detections are considered, reducing noise.

6.4 Frame Optimization

Frames were resized to an optimal resolution to maintain a balance between processing speed and detection accuracy.

6.5 Segmentation Integration

Body segmentation was implemented to separate the human subject from the background. This improves visual clarity and helps in focusing on the subject.

6.6 Performance Monitoring

FPS (Frames Per Second) was calculated and displayed to monitor the performance of the system in real time.

7. Results and Observations

The improved pose detection system showed the following results:

- Accurate detection of body landmarks in real time
- Stable skeleton tracking with reduced jitter
- Improved visual clarity due to background segmentation
- Efficient performance with real-time processing
- Enhanced user experience with FPS display

8. Limitations

Despite improvements, some limitations were observed:

- Detection accuracy depends on lighting conditions
- Performance decreases in low-light environments
- Partial detection occurs if the full body is not visible
- Segmentation may produce errors in complex backgrounds

9. Conclusion

The improved pose detection system successfully demonstrates how artificial intelligence can be used to track human body movements in real time. By optimizing model parameters, enabling smoothing, integrating segmentation, and monitoring performance, the system became more accurate, stable, and efficient.

This implementation forms a strong foundation for advanced applications such as virtual try-on systems, gesture recognition, fitness tracking, and AI-based fashion recommendation systems.